

Project meeting 1: Our research project

Minidiss 26/27 with Dr. Elizabeth Pankratz

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1 Warm-up

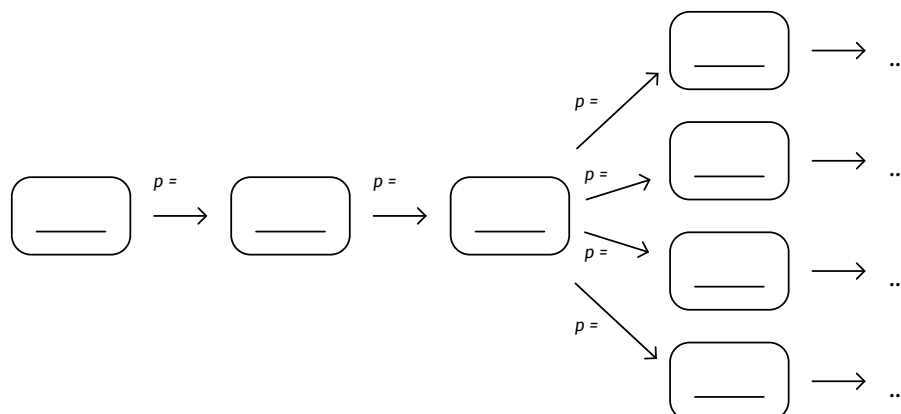
- Make a name card that includes:
 - Your first name
 - The pronouns you want me to use for you
 - Your UUN (looks like s1234567)
- Introduce yourselves!
 - What's your name?
 - What was the best thing you did/saw/experienced this summer? (do your best to remember everybody's thing)
 - What do you like about studying psychology?
- You can find all kinds of information about the project we'll do together on the project website:

`elizabethpankratz.github.io/minidiss2627/`

- I'll update this website with handouts, links, and key tasks for every meeting.

2 What is statistical learning?

- First: Let's do an experiment!
- This experiment uses an artificial "language" containing four three-syllable "words", repeated and randomly shuffled:
 1. _____
 2. _____
 3. _____
 4. _____
- In this "language"..
 - After the syllable *bu*, what syllables can come next?
 - After the syllable *ka*, what syllables can come next?
 - After the syllable *di*, what syllables can come next?
- The probability that a specific syllable will be followed by another specific syllable = **transitional probability (TP)**.
- Let's fill in the following schematic that shows part of the distributional structure of this language:



3 Why might autistic people do statistical learning differently than allistic (non-autistic) people?

- You've maybe heard that autistic people notice smaller details that allistic people might miss, and that autistic people function best with routine/predictability.
- Here's how autism researchers talk about these traits:
 - “attention to detail” = priority for local processing over global processing (Mottron et al., 2006)
 - “functioning better with routine” = preference for predictable input (Van De Cruys et al., 2014)
- These two traits make opposing predictions about how autistic people might deal with transitional probabilities that are lower than 1 in a statistical learning task.
 - Autistic people might be better than allistic people at learning low-TP sequences because of their stronger local processing.
 - Autistic people might be worse than allistic people at learning low-TP sequences because they're more disrupted by the lack of predictability.
- Or maybe some autistic people process things one way, and other autistic people process things a different way?
- Our goal this year is to find out!

4 Big picture: What to expect from our project this year

- We will work through the whole research process together, step by step.
- Our project meetings will nearly all be hands-on and interactive, like today.
 - I'll randomise the seating arrangement at the beginning of every meeting, so that you have a chance to sit with and talk to different people.
 - I'll lead you through structured collaborative activities, so that you have a chance to play with the ideas yourselves.
 - Bring a pen to each project meeting—I'll nearly always have paper handouts for you to scribble on.
- A research project has many different parts. **Before our next meeting, please sign up for the parts that you would like to work on.**
 - You'll find the spreadsheet linked on the project website page for Meeting 1.
 - Why does everyone sign up for multiple tasks? So that you can share your experience from one task with your collaborators on a different task, and they can share their experience with you.
 - The goal is for everyone to have a reasonable overview of the whole project.
- Please let me know in advance if you can't attend one of our project meetings!

References and recommended reading

- Isbilen, E. S., & Christiansen, M. H. (2022). Statistical Learning of Language: A Meta-Analysis Into 25 Years of Research. *Cognitive Science*, 46(9), 1–35. <https://doi.org/10.1111/cogs.13198>
- Mottron, L., Dawson, M., Soulières, I., Hubert, B., & Burack, J. (2006). Enhanced Perceptual Functioning in Autism: An Update, and Eight Principles of Autistic Perception. *Journal of Autism and Developmental Disorders*, 36(1), 27–43. <https://doi.org/10.1007/s10803-005-0040-7>
- Saffran, J. R., Newport, E. L., & Aslin, R. N. (1996). Word segmentation: The role of distributional cues. *Journal of Memory and Language*, 35(4), 606–621. <https://doi.org/10.1006/jmla.1996.0032>
- Van De Cruys, S., Evers, K., Van Der Hallen, R., Van Eylen, L., Boets, B., de Wit, L., & Wagemans, J. (2014). Precise minds in uncertain worlds: Predictive coding in autism. *Psychological Review*, 121(4), 649–675. <https://doi.org/10.1037/a0037665>